



National Institute of
Diabetes and Digestive
and Kidney Diseases

What Is Diabetic Neuropathy?

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Diabetic neuropathy is nerve damage that is caused by [diabetes](#).

Nerves are bundles of special tissues that carry signals between your brain and other parts of your body. The signals

- send information about how things feel
- move your body parts
- control body functions such as [digestion](#)

What are the different types of diabetic neuropathy?

Types of diabetic neuropathy include the following:

Peripheral neuropathy

[Peripheral neuropathy](#) is nerve damage that typically affects the feet and legs and sometimes affects the hands and arms.

Autonomic neuropathy

[Autonomic neuropathy](#) is damage to nerves that control your internal organs. Autonomic neuropathy can lead to problems with your heart rate and blood pressure, digestive system, bladder, sex organs, sweat glands, eyes, and ability to sense [hypoglycemia](#).

Focal neuropathies

[Focal neuropathies](#) are conditions in which you typically have damage to single nerves, most often in your hand, head, torso, and leg.

Proximal neuropathy

[Proximal neuropathy](#) is a rare and disabling type of nerve damage in your hip, buttock, or thigh. This type of nerve damage typically affects one side of your body and may rarely spread to the other side. Proximal neuropathy often causes severe pain and may lead to significant weight loss.

Who is most likely to get diabetic neuropathy?

If you have diabetes, your chance of developing nerve damage caused by diabetes increases the older you get and the longer you have diabetes. Managing your diabetes is an important part of preventing health problems such as diabetic neuropathy.



Managing your diabetes is an important part of preventing health problems such as diabetic neuropathy.

You are also more likely to develop nerve damage if you have diabetes and

- are overweight
- have [high blood pressure](#)
- have high [cholesterol](#)
- have advanced **kidney disease**
- drink too many alcoholic drinks
- smoke

Research also suggests that certain [genes](#) may make people more likely to develop diabetic neuropathy.

What causes diabetic neuropathy?

Over time, high blood glucose levels, also called blood sugar, and high levels of fats, such as [triglycerides](#), in the blood from diabetes can damage your nerves. High blood glucose levels can also damage the small [blood vessels](#) that nourish your nerves with oxygen and nutrients. Without enough oxygen and nutrients, your nerves cannot function well.

How common is diabetic neuropathy?

Although different types of diabetic neuropathy can affect people who have diabetes, research suggests that up to one-half of people with diabetes have peripheral neuropathy.^{1,2} More than 30 percent of people with diabetes have autonomic neuropathy.²

The most common type of focal neuropathy is [carpal tunnel syndrome](#) [NIH](#), in which a nerve in your wrist is compressed. Although less than 10 percent of people with diabetes feel symptoms of carpal tunnel syndrome, about 25 percent of people with diabetes have some nerve compression at the wrist.²

Other focal neuropathies and proximal neuropathy are less common.

What are the symptoms of diabetic neuropathy?

Your symptoms depend on which type of diabetic neuropathy you have. In peripheral neuropathy, some people may have a loss of sensation in their feet, while others may have burning or shooting pain in their lower legs. Most nerve damage develops over many years, and some people may not notice symptoms of mild nerve damage for a long time. In some people, severe pain begins suddenly.

What problems does diabetic neuropathy cause?

Peripheral neuropathy can lead to [foot complications](#), such as sores, ulcers, and infections, because nerve damage can make you lose feeling in your feet. As a result, you may not notice that your shoes are causing a sore or that you have injured your feet. Nerve damage can also cause problems with balance and coordination, leading to falls and fractures.

These problems may make it difficult for you to get around easily, causing you to lose some of your independence. In some people with diabetes, nerve damage causes chronic pain, which can lead to anxiety and [depression](#) [NIH](#).

Autonomic neuropathy can cause [problems with how your organs work](#), including problems with your heart rate and blood pressure, digestion, [urination](#), and ability to sense when you have low blood glucose.

How can I prevent diabetic neuropathy?

To prevent diabetic neuropathy, it is important to **manage your diabetes** by managing your blood glucose, blood pressure, and cholesterol levels.

You should also take the following steps to help prevent diabetes-related nerve damage:

- be physically active
- follow your diabetes meal plan
- get help to [quit smoking](#) 
- limit alcoholic drinks to no more than one drink per day for women and no more than two drinks per day for men
- take any **diabetes medicines** and other medicines your doctor prescribes



To help prevent diabetes-related nerve damage, get help to quit smoking.

How can I prevent diabetic neuropathy from getting worse?

If you have diabetic neuropathy, you should manage your diabetes, which means managing your blood glucose, blood pressure, cholesterol levels, and weight to keep nerve damage from getting worse.

Foot care is very important for all people with diabetes, and it's even more important if you have peripheral neuropathy. Check your feet for problems every day, and [take good care of your feet](#). See your doctor for a neurological exam and a [foot exam](#) at least once a year—more often if you have foot problems.

References

- [1] Pop-Busui R, Boulton AJ, Feldman EL, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care*. 2017;40(1):136–154.
- [2] Izenberg A, Perkins BA, Bril V. Diabetic neuropathies. *Seminars in Neurology*. 2015;35(4):424–430.

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